
Software Requirements Specification

for

Personal Finance Tracker

Version 1.0 approved

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1. Introduction

In today's fast-paced world, managing personal finances efficiently is crucial for financial stability and growth. A Personal Finances Tracking System plays a vital role in helping individuals control expenses, plan budgets, track savings, and make informed financial decisions.

1.1 Purpose

The Personal Finances Tracking System (PFTS) is designed to help individuals efficiently manage their financial transactions, including income, expenses, budgeting, and savings. The system provides an intuitive platform for users to track their financial activities, generate reports, and plan their budgets for better financial control.

1.2 Document Conventions

The SRS document for the Personal Finance Tracker project follows a structured format to ensure consistency, readability, and clarity. The following conventions will be maintained throughout the document:

1. Text Formatting Conventions

- **Bold Text:** Used for section headings and important terms.
- *Italic Text:* Used for emphasis and project-specific terms.
- `Monospace Text:` Used for code snippets, API references, and file paths.

2. Numbering and Organization

- Sections and subsections are hierarchically numbered (e.g., 1, 1.1, 1.1.1).
- Lists and bullet points are used to enhance readability and organization.

3. Terminology & Definitions

- Common software engineering terms (e.g., "API," "Database," "User Interface") will follow standard definitions.
- Project-specific terms (e.g., "Expense Category," "Financial Insights") will be defined in the Glossary section.

4. References and Citations

- Any external references (books, research papers, web sources) will be cited in IEEE format in the "References" section.

5. Figures and Diagrams

- Diagrams (UML, flowcharts, wireframes) will be numbered and labeled (e.g.,

Figure 1: System Architecture).

- Tables will be labeled accordingly (e.g., Table 1: Feature Comparison).

6. Requirements Specification Format

- Functional and non-functional requirements will be documented using a structured format (e.g., Requirement ID, Description, Priority, Status).
- Each requirement will follow IEEE 830 standards for clarity and completeness.

7. Assumptions and Dependencies

- Any project-related assumptions (e.g., "Users have basic financial knowledge") will be explicitly stated.
- Software dependencies (e.g., Python, Flask, Power BI, yfinance) will be listed in the "System Dependencies" section.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Developers:** To understand system functionality and implementation details.
- **Project Managers:** To oversee development and ensure compliance with requirements.
- **Testers:** To verify system performance and correctness.
- **End Users:** To understand system features and limitations.

1.4 Product Scope

The Personal Finances Tracking System is a web-based and mobile application designed to provide users with:

- Income and Expense Tracking: Record financial transactions.

- **Budget Planning:** Set monthly budgets for different expense categories.
- **Savings Management:** Track and manage personal savings.
- **Report Generation:** Generate graphical and tabular reports for financial insights.
- **Notification System:** Reminders for bill payments and budget alerts.
- **Data Security:** Ensuring encrypted storage of financial data.

The system aims to provide a user-friendly interface with a focus on security, accuracy, and efficiency in financial management.

1.5 References

- **IEEE 830-1998 Standard** – Guidelines for writing structured Software Requirements Specifications.

- **Sommerville, I. (2015)** – Software Engineering principles and best practices.
- **McKinney, W. (2017)** – Python for Data Analysis, covering Pandas, NumPy, and Jupyter.
- **Power BI Documentation** – Official guide for data visualization and dashboard creation.
- **Tableau User Guide** – Documentation for implementing financial data visualization.

2. Overall Description

2.1 Product Perspective

The Personal Finances Tracking System is a standalone system that interacts with cloud-based storage services for data backup and synchronization across multiple devices. It is designed for personal financial tracking and does not integrate with banking institutions for real-time transactions.

2.2 Product Functions

The system provides the following core functionalities:

1. **User Authentication:** Secure login and account management.
2. **Transaction Management:** Add, income and expense transactions.
3. **Budgeting System:** Set limits for different expense categories.
4. **Savings Tracker:** Monitor savings progress.
5. **Reports and Analytics:** Graphical representation of financial data.

2.3 User Classes and Characteristics

- **Regular Users:** Individuals using the system for personal finance tracking.
- **Administrator:** Manages system maintenance, security updates, and bug fixes.

2.4 Operating Environment

- **Web Version:** Accessible via browsers like Chrome, Firefox, Edge.
- **Mobile Application:** Supports Android (6.0 and above) and iOS (12.0 and above).
- **Database:** PHP for data storage.
- **Backend:** Python
- **Frontend:** HTML, CSS and JavaScript.

2.5 Design and Implementation Constraints

- Must support multiple platforms (web and mobile).
- Secure user authentication required (OAuth or JWT-based).
- Cloud storage integration for data backup.

- Performance optimization for faster loading times.

2.6 User Documentation

- **Installation & Setup:** Install Python 3.8+, required libraries (flask, pandas, matplotlib), and run the application using `python app.py` in a web browser at `http://localhost:5000/`.
- **Features & Usage:** Users can track income and expenses, view real-time financial insights, and visualize trends through interactive dashboards (Power BI/Tableau).

2.7 Assumptions and Dependencies

- Users have stable internet access for cloud-based features.
- System depends on third-party APIs for notification services.
- Regular security updates are necessary for system integrity.

3. External Interface Requirements

3.1 User Interfaces

- **Login & Registration** – Secure user authentication with email, password, and recovery options.
- **Dashboard** – Overview of income, expenses, savings, and financial insights with visual graphs.
- **Income & Expense Entry** – Add, edit, delete, and categorize financial transactions.
- **Reports & Analytics** – Power BI/Tableau integration for trend analysis and custom reports.
- **Settings & Profile** – Manage user details, notifications, and export financial data.

3.2 Hardware Interfaces

- **Database Server** – PHP databases, opens through phpMyAdmin localhost server.
- **Networking Hardware** – Requires Wi-Fi or Ethernet connectivity for dashboard updates.
- **Peripheral Devices (Optional)** – Supports printers for financial reports, biometric devices for secure authentication, and external storage for data backup.
- **Cloud Integration (Optional)** – Can be hosted on cloud servers for scalability, enabling remote access and secure financial data storage.

3.3 Software Interfaces

- **Operating System** – Compatible with Windows, macOS, and Linux for local deployment and

cloud-based access.

- **Backend Framework** – Built using Flask/Django for handling business logic, user authentication, and database management.
- **Database Management System (DBMS)** – Supports SQL (PostgreSQL, MySQL, SQLite) or NoSQL (MongoDB) for storing user transactions and financial data.
- **Frontend Interface** – Uses HTML, CSS, JavaScript (React.js/Angular.js) for an interactive and user-friendly UI.
- **API Integration** – Connects with yFinance API for real-time stock data, Power BI/Tableau APIs for financial analytics, and payment gateways (Stripe/PayPal) for transaction tracking.

3.4 Communications Interfaces

- **HTTP/HTTPS Protocols** – Ensures secure communication between the client (frontend) and server (backend) using RESTful APIs.
- **Database Communication** – Uses SQL (PostgreSQL/MySQL/SQLite) or NoSQL (MongoDB) for handling financial data transactions securely.
- **API Communication** – Integrates with external APIs like yFinance (financial data retrieval), Power BI/Tableau (data visualization), and payment gateways (Stripe/PayPal).
- **Email & Notification Services** – Uses SMTP, Firebase, or Twilio for email alerts, SMS notifications, and budget reminders.

4. System Features

4.1 User Authentication & Security

Description and Priority:

- Provides a secure login and registration system with encrypted password storage.
- Ensures role-based access (admin, standard user).
- Priority: High

Stimulus/Response Sequences:

- User enters login credentials → System verifies and grants access.
- Incorrect login attempt → System prompts an error message.

Functional Requirements:

- REQ-1: The system shall allow users to register and log in securely.
- REQ-2: The system shall encrypt all stored passwords.
- REQ-3: Users shall be able to reset their passwords via email verification.

4.2 Income & Expense Management

Description and Priority:

- Allows users to record, categorize, and track their income and expenses.
- Provides filtering options based on date, category, and transaction type.
- Priority: High

Stimulus/Response Sequences:

- User adds a transaction → System updates the financial summary.
- User deletes a transaction → System removes it and recalculates balances.

Functional Requirements:

- REQ-4: The system shall allow users to add, edit, and delete transactions.
- REQ-5: The system shall support categorization of transactions (e.g., Rent, Food, Investment).
- REQ-6: Users shall be able to filter and search for transactions based on date and category.

4.3 Financial Data Visualization

Description and Priority:

- Provides interactive dashboards displaying financial summaries, expense trends, and savings insights.
- Integrates with Power BI/Tableau for data visualization.
- Priority: High

Stimulus/Response Sequences:

- User selects a date range → System generates charts and graphs based on transaction data.
- User applies filters → System updates the displayed financial trends accordingly.

Functional Requirements:

- REQ-7: The system shall generate interactive reports for income and expenses.
- REQ-8: Users shall be able to export financial reports in CSV/PDF formats.
- REQ-9: The system shall integrate with Power BI/Tableau for enhanced visualization.

4.4 Budgeting & Expense Prediction

Description and Priority:

- Uses data science techniques to predict future expenses and savings based on transaction history.
- Provides budgeting suggestions based on spending patterns.
- Priority: Medium

Stimulus/Response Sequences:

- User enters monthly income → System recommends a budget allocation.
- User exceeds budget limit → System sends a spending alert.

Functional Requirements:

- REQ-10: The system shall analyze past spending to predict future expenses.
- REQ-11: Users shall receive budget recommendations based on income and expenses.
- REQ-12: The system shall send alerts when expenses exceed predefined limits.

4.5 Reports & Data Export**Description and Priority:**

- Generates financial reports for analysis and tax filing.
- Allows users to export reports in multiple formats (CSV, PDF, Excel).
- Priority: Low

Stimulus/Response Sequences:

- User requests an expense report → System generates and downloads the report.

Functional Requirements:

- REQ-16: The system shall allow users to generate custom financial reports.
- REQ-17: Users shall be able to export financial data in multiple formats.

5. Other Nonfunctional Requirements**5.1 Performance Requirements:**

The system shall ensure dashboard loading within 3 seconds and report generation within 5 seconds, supporting at least 1,000 concurrent users without performance degradation (NFR-1, NFR-2, NFR-3).

5.2 Safety Requirements:

The system shall maintain data integrity with automatic daily backups and prevent financial data loss or corruption, ensuring transactional reliability (NFR-4, NFR-5).

5.3 Security Requirements:

User data shall be protected using AES-256 encryption, multi-factor authentication (MFA), and session expiration after 30 minutes of inactivity, ensuring compliance with financial data privacy laws (NFR-6, NFR-7, NFR-8).

5.4 Software Quality Attributes:

The system shall be responsive, mobile-friendly, and scalable, with a modular architecture that allows easy maintenance and feature upgrades, ensuring long-term reliability (NFR-9, NFR-10, NFR-11).

5.5 Business Rules:

The system shall ensure accurate expense categorization, customizable budgeting rules, and automated financial tracking, allowing users to set personalized spending limits and savings goals (NFR-12, NFR-13).

6. Other Requirements

The system shall comply with financial data privacy regulations such as GDPR and PCI DSS to ensure user data security and confidentiality (NFR-14). It shall support multi-currency and multi-language options to cater to international users (NFR-15). Additionally, the system shall be compatible with third-party financial APIs for seamless data integration and allow export of reports in CSV, PDF, and Excel formats for external use (NFR-16, NFR-17). The application shall also provide audit logs for transaction tracking and role-based access control to enhance administrative security.

Appendix A: Glossary

- **API (Application Programming Interface):** A set of protocols and tools that allow different software applications to communicate with each other.
- **Authentication:** The process of verifying a user's identity before granting access to the system.
- **Budgeting:** The process of planning and tracking income and expenses to manage finances effectively.
- **Dashboard:** A graphical interface that provides an overview of financial data using charts and summaries.
- **Data Encryption:** A security measure that converts sensitive information into an unreadable format to protect it from unauthorized access.
- **Database:** A structured collection of data used to store and manage financial transactions and user information.
- **Expense Tracking:** The process of recording and categorizing spending activities to monitor financial health.

- **GDPR (General Data Protection Regulation):** A legal framework that sets guidelines for collecting and processing personal information from individuals.
- **Multi-Factor Authentication (MFA):** A security mechanism that requires users to provide multiple forms of verification before accessing their accounts.
- **PCI DSS (Payment Card Industry Data Security Standard):** A set of security standards designed to ensure the secure handling of credit card information.
- **Power BI:** A data visualization tool used to create financial reports and dashboards.
- **Report Generation:** The process of creating financial summaries and insights in formats like PDF, CSV, or Excel.
- **Stock Market Tracking:** Monitoring real-time financial market data using APIs like yFinance.
- **Transaction:** A financial activity involving income, expenses, or investments recorded in the system.
- **User Role Management:** A feature that allows different levels of access for administrators, standard users, and guests.

Appendix B: Analysis Models

The Personal Finance Tracker system follows a structured analysis model to define its functionality, data flow, and user interactions. The models include Use Case Diagrams, Data Flow Diagrams (DFD), Entity-Relationship Diagrams (ERD), and System Architecture Diagrams.

Use Case Diagram

- Represents the interactions between users (admin, registered users) and the system.
- Key use cases: User Authentication, Expense Tracking, Budgeting, Report Generation, Investment Monitoring, and Dashboard Visualization.

Data Flow Diagram (DFD)

- Shows how financial data moves through the system, from user input (income/expenses) to database storage and report generation.
- DFD Level 0: High-level system overview.
- DFD Level 1: Detailed data interactions, including APIs for stock data and visualization tools.

Entity-Relationship Diagram (ERD)

- Defines the relationships between database entities:
- Users (User_ID, Name, Email, Role)

- Transactions (Transaction_ID, User_ID, Amount, Category, Date)
- Budgets (Budget_ID, User_ID, Category, Limit)
- Investments (Investment_ID, User_ID, Stock_Name, Shares, Price, Date)

System Architecture Diagram

- Illustrates the 3-tier architecture (Frontend, Backend, and Database layers).
- Technologies: Frontend (React.js/HTML/CSS), Backend (Flask/Django), Database (PostgreSQL/MongoDB), APIs (yFinance, Power BI, Payment Gateway).

Appendix C: To Be Determined List

The following items are yet to be finalized for the Personal Finance Tracker project:

- **TBD-1:** Selection of the final backend framework – Flask API.
- **TBD-2:** Choice of database system – PHP.
- **TBD-3:** Decision on cloud hosting provider – AWS, Google Cloud, or Azure.
- **TBD-4:** Implementation of advanced AI-driven financial insights and recommendations.